Occupational Health and Safety-1

Fire safety and Prevention

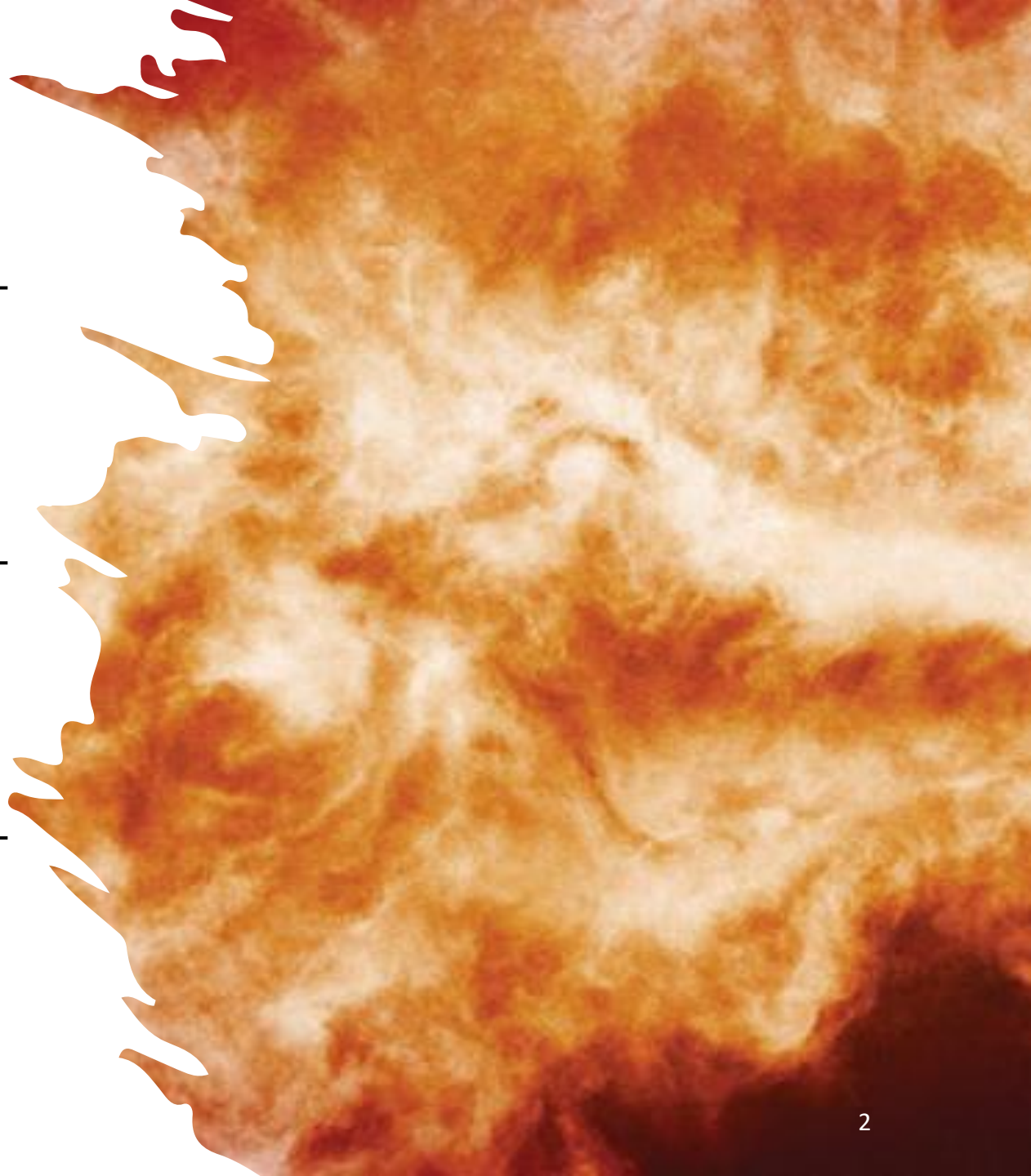
Fire, explosion, flash

FIRE

Fire Science: it is the rapid oxidation of a material in the exothermic chemical process of combustion, releasing heat, light & various reaction products.

Workplace Awareness: Fire prevention in the workplace starts out with knowledge of knowing different types of fires & how to control them in the case of emergency.

Fire Types and Extinguishers: There are four different types of fires each of which have different fire extinguishers to be used.



CHARACTERISTICS OF FIRE

Fire can cause large losses (human & materials)

Chemical reaction requires three components : Fuel , Oxygen, Ignition source [Fire triangle].

Ignition can be caused by convection , conduction & radiation .

Several type of combustion process depending on the combustion rate.

- A) Slow combustion (no flames , very low temp)
- B) Normal combustion (controlled combustion rate)
- C) Explosion (very quickly)
- D) Detonation (very quickly & propagates very quickly)

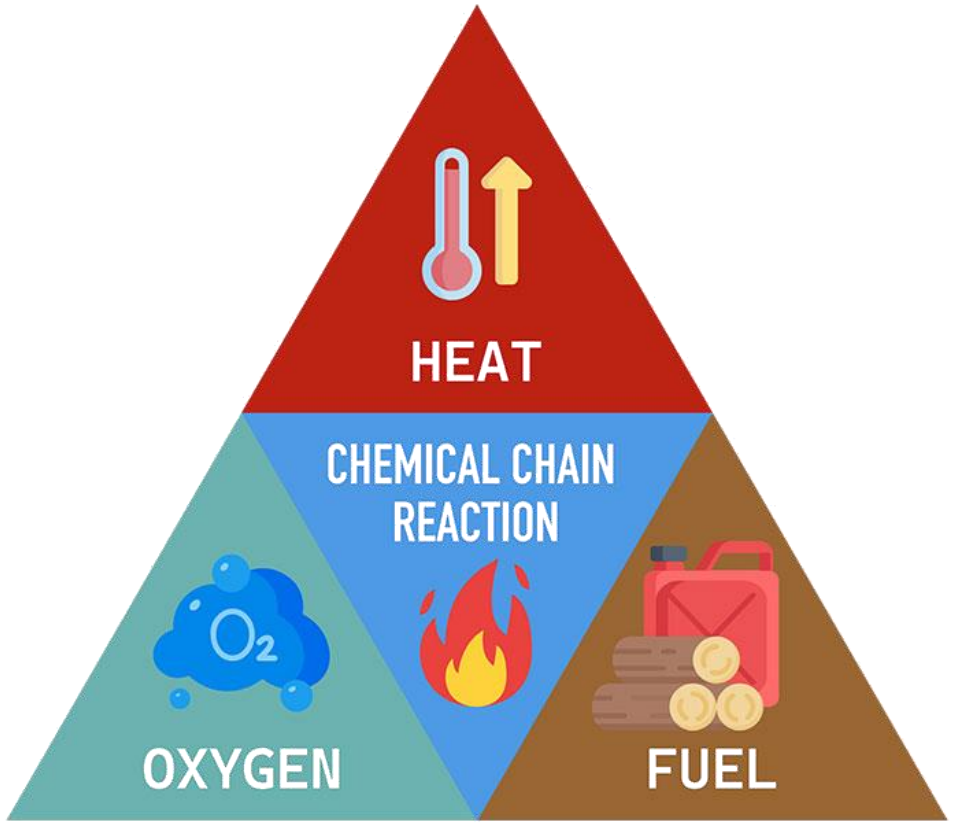
What is the Fire triangle?

Components of Fire: Oxygen, Heat, Fuel.

Interaction: Removing any one component prevents or extinguishes a fire.

Importance in Safety: Knowing the fire triangle helps in identifying and controlling fire risks effectively.

Example: A fire extinguisher removes oxygen or reduces heat to suppress the fire.



Oxygen

- It is a colorless, odorless and tasteless gas that is present at a rate of 21% in the air.
- It liquefies at -183°C and is usually stored as a liquid.
- It does not burn itself, but it performs the combustion.
- The ratio of oxygen in air is 21%. A fire needs at least 16% oxygen to survive.
- Smoldering, on the other hand, occurs with 3% oxygen in the air.

Sources of Fuel

Flammable Liquids

- Ethanol, Acetone and a whole range of other solvents

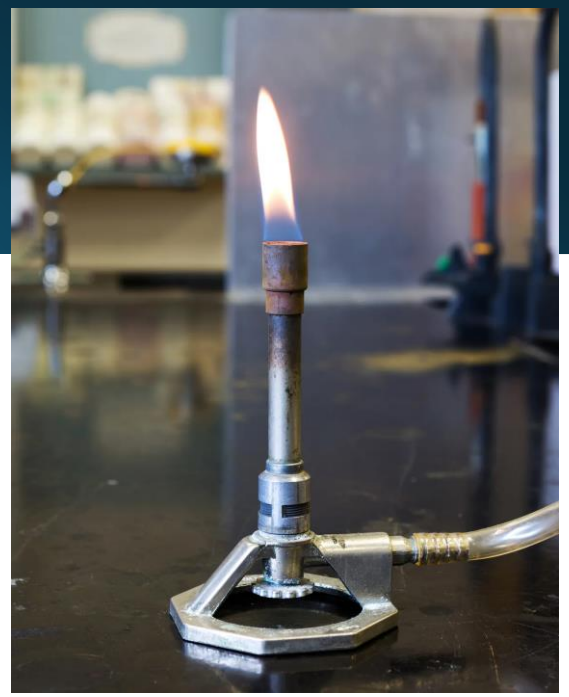
Flammable Gases

- Hydrogen, Acetylene

Carbonaceous Materials

- Wood (benches, furniture), Paper (tissues, books and office paperwork), cardboard (boxes), foam and plastics (packing materials, contents of upholstered furniture), textiles (clothing, curtains), animal bedding materials.

Sources of Ignition



Naked Flames

- Bunsen burners, candles.

Hot Surfaces

- Hotplates, hot machine surfaces, cooking appliances

Faulty Electrical Equipment

- Extension leads are a particular hazard and can caused fires

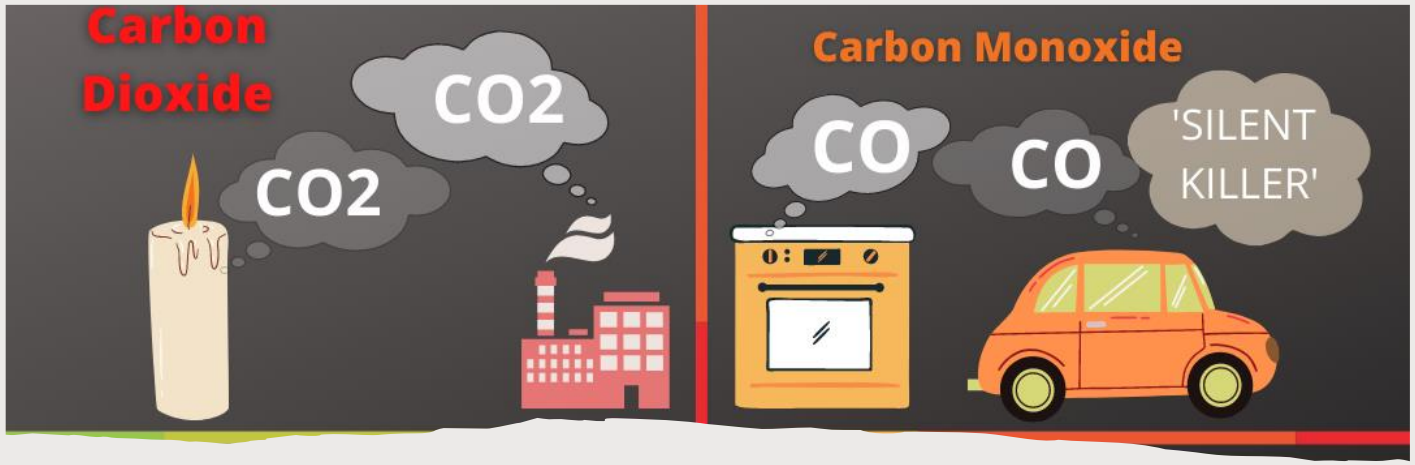
Heaters

- Radiant or other types of heaters

Safety Tip: Regular maintenance of electrical devices reduces ignition risks.

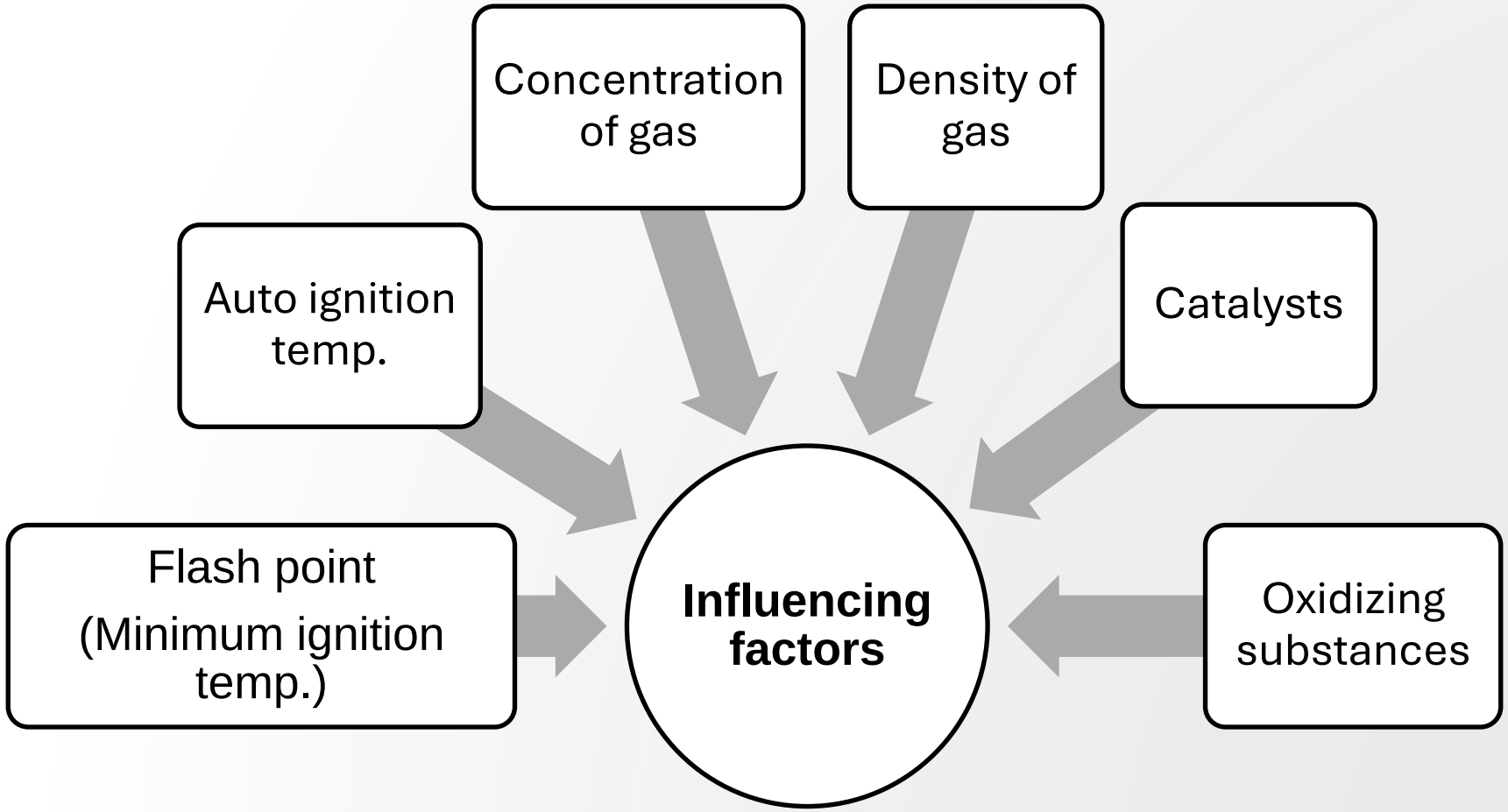
By-Product

- **Carbon Monoxide**-Incomplete combustion-poisonous
- **Carbon Dioxide**-Complete combustion-displaces oxygen
- **Fuel Particulate Matter** – unburned fuels – carcinogen (cancer causing)
- **Smoke** – unburned materials – can contain all of the above



Prevention: Ensure adequate ventilation and early detection systems

Factors influencing fire:



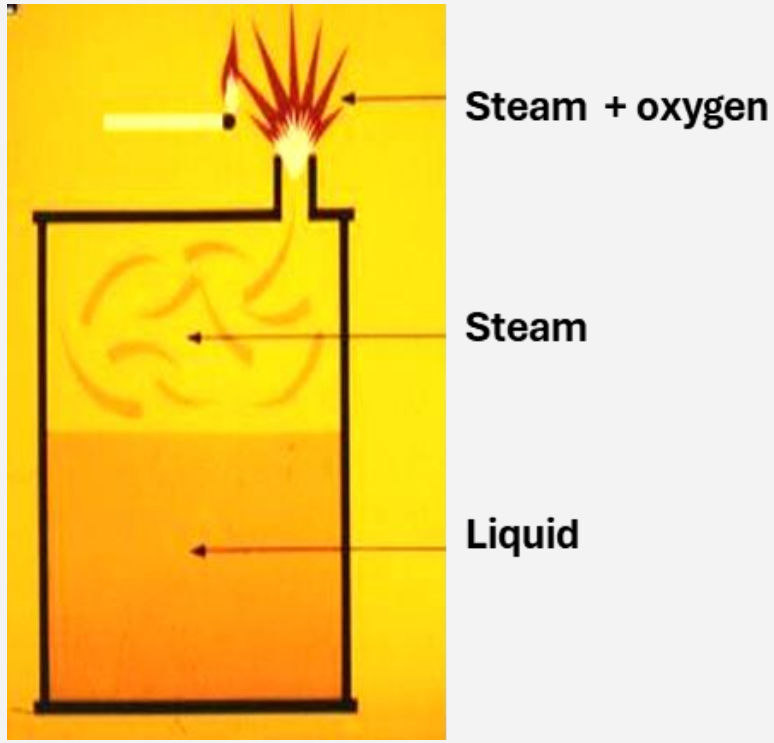
Flash Point

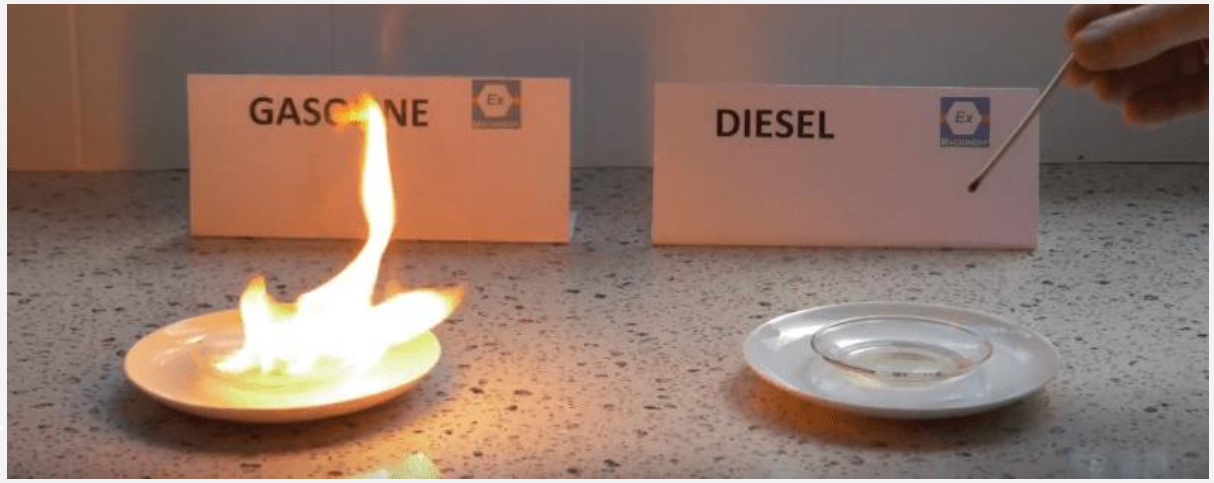
Definition: The lowest temperature at which a flammable liquid forms an ignitable air mixture.

Significance: Essential for classifying and storing flammable liquids.

Examples: Gasoline has a flash point of -43°C, while diesel is around 52°C.

Safety: Store liquids below their flash points to prevent accidental ignition.





Ignition Point

- The temperature at which solid materials begin to emit flammable gases with heat energy is the **"Ignition point"**.
- Combustion occurs in the gas phase.
- The greater the surface of the substance, the easier it is to ignite

Explosion

- It is a sudden oxidation or decomposition reaction that occurs with a sudden change in temperature, pressure, or both.

- In explosive organic substances; The chain combustion reaction is completed very quickly (1/700 to 1/1.000 sec) due to the presence of O_2 required for the combustion of C and H atoms in its own molecular structure.

Explosion

- The effect of the explosive occurs when the air, which expands with the heat energy released, expands by spreading shock waves around it.

DEFLEGRATION: It is an explosion that propagates at subsonic speed.

DETONATION: An explosion characterized by a shock wave and propagating at supersonic speed.

Explosion Limits (fuel-to-air mixtures)

If it is more than the Upper Explosion Limit: “Rich Mixture”

upper explosion limit

Explosive Zone

Lower explosion limit

If it is less than the Lower Explosion Limit: “Lean Mixture”

Types of explosion

Natural explosion :

- These are the explosions that occur in nature such as volcanic eruptions & explosions , during forest fires from the igniting of oils in three tops.

Chemical explosion :

- These are the most common explosion that occur, mostly in the areas that use it during the job operation such as ammunition testing areas.
- These type including explosion such as gun powder , TNT, & dynamite.

Types of explosion

Fukushima disaster

Nuclear explosion:

- This is an explosion derived from fission & fusion in a nuclear reaction
- nuclear explosion is powerful enough to destroy an entire city, making them deadly hazard.

Electrical explosion:

- Excessive magnetic pressure within an ultra – strong electromagnet can cause a magnetic explosion .
- A lightning bolt is another form of electric explosion .

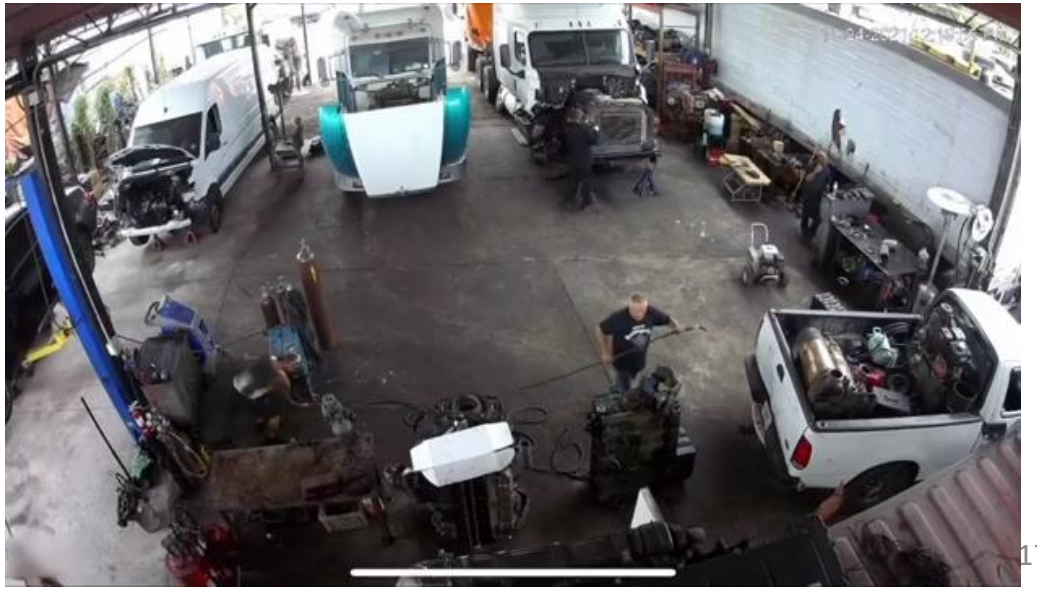
Types of explosion

Mechanical explosion:

- These are completely physical, mechanical, occur when a tightly sealed container under pressure encounter a break , at which it release its gases or liquids rapidly causing an explosion.

Explosions Terminology : Mechanical Explosion

Gas Tank Explodes



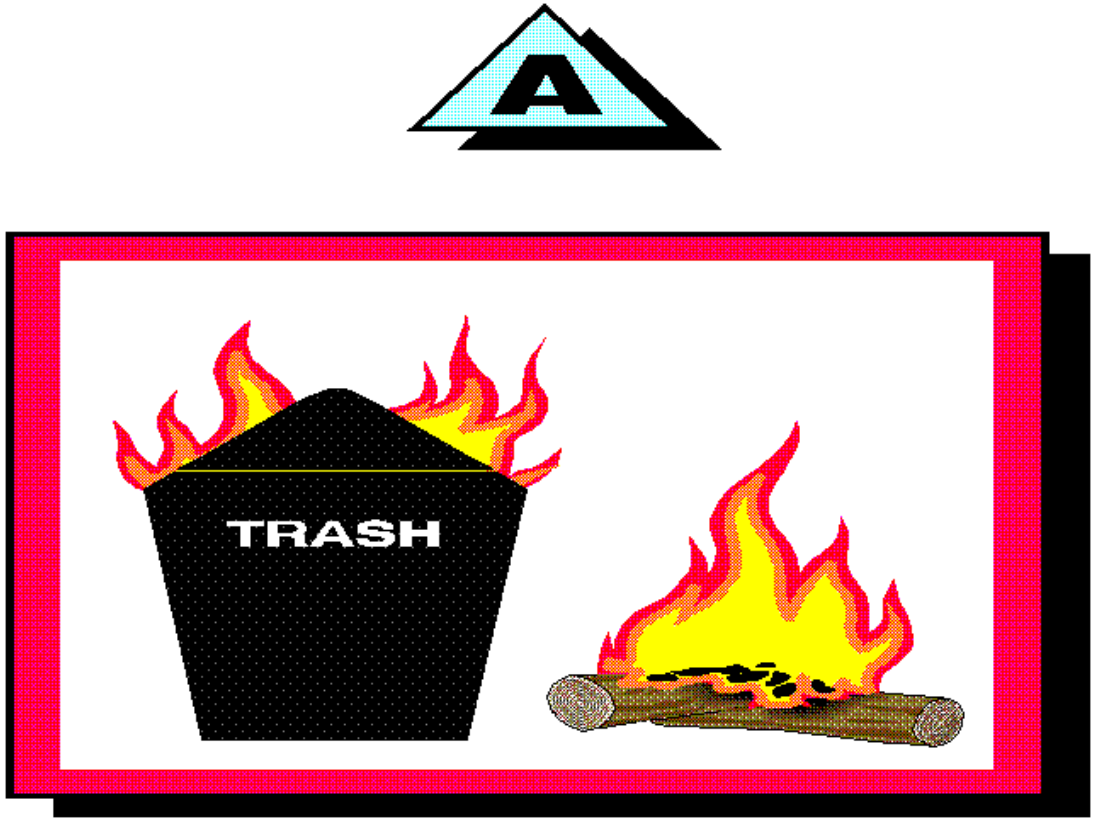
What are the 4 types of fire?

1. Class A : Burning flammable solids
2. Class B : Burning flammable liquids
3. Class C : Fires burn flammable gases
4. Class D : Burning flammable metals

Class A - Materials

Solids

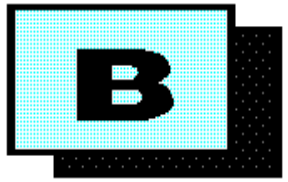
- Wood
- Paper
- Plastic Rubber

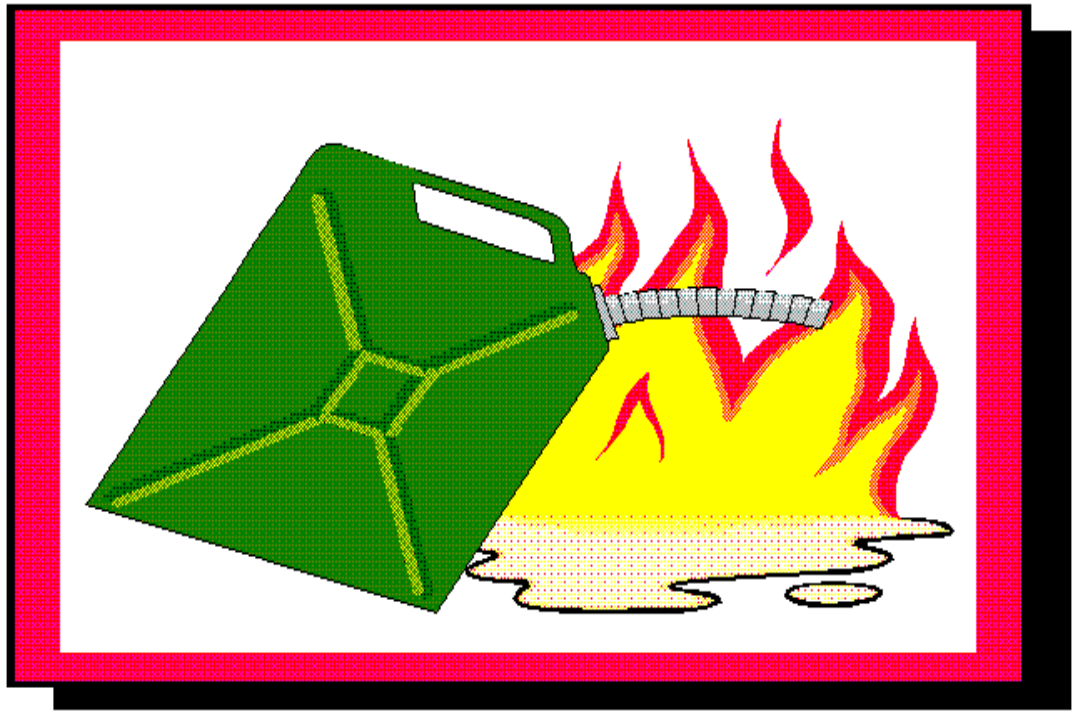


Class B - Materials

Liquids

- Gasoline, Oil, Diesel fuel
- Greases, Hydraulic fluid





Class C - Materials

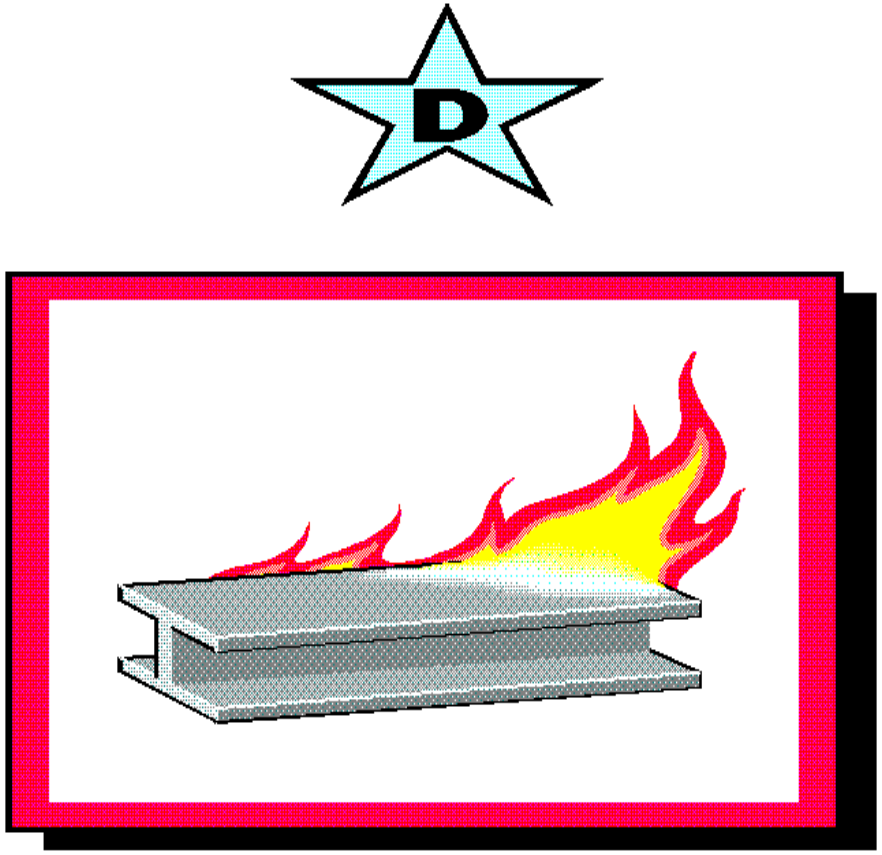
Flammable Gases

Flammable gases escaped from their containers (cylinders or pipes), usually in pressurized forms. They can either ignite quickly to form a flash or a jet fire (if it is pressurized) or they can expand into the space to ignite later into a vapor cloud explosion. To extinguish C fires, it is usually necessary to cut out gas first.

Class D - Materials

Combustible Metals

- Magnesium
- Titanium
- Zirconium
- Sodium
- Potassium
- Lithium
- Calcium
- Zinc



Class of fuels on extinguisher

Type(s) of fire it can put out

How much fire a "lay person" can put out

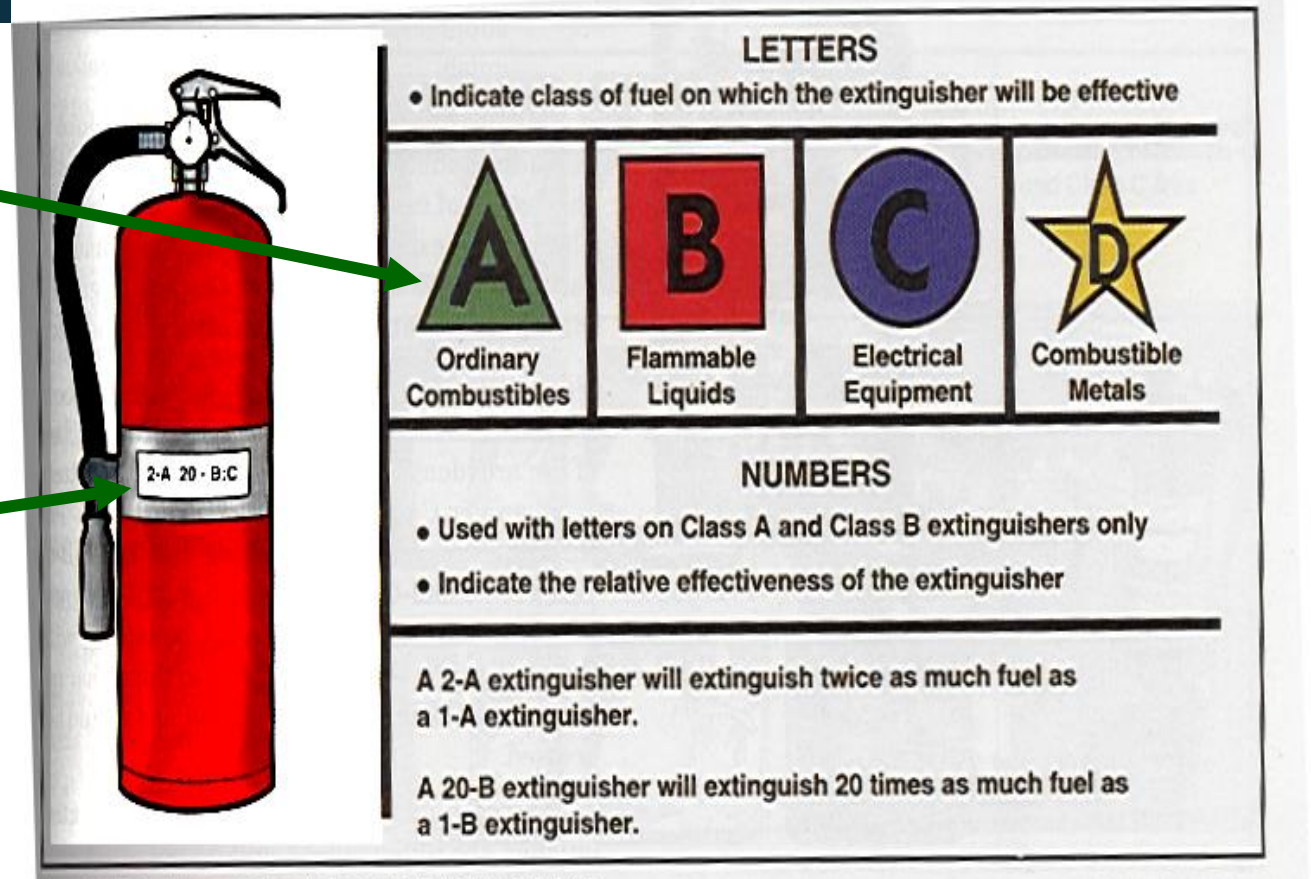


Figure 5.24 Extinguishers are classified according to their intended use.

- A properly trained person can extinguish 2 to 3 times the amount listed on the rating.

When are fire extinguishers to be examined?

1. Fire extinguishers shall be inspected visually at least once a month to determine that they are fully charged and operable.

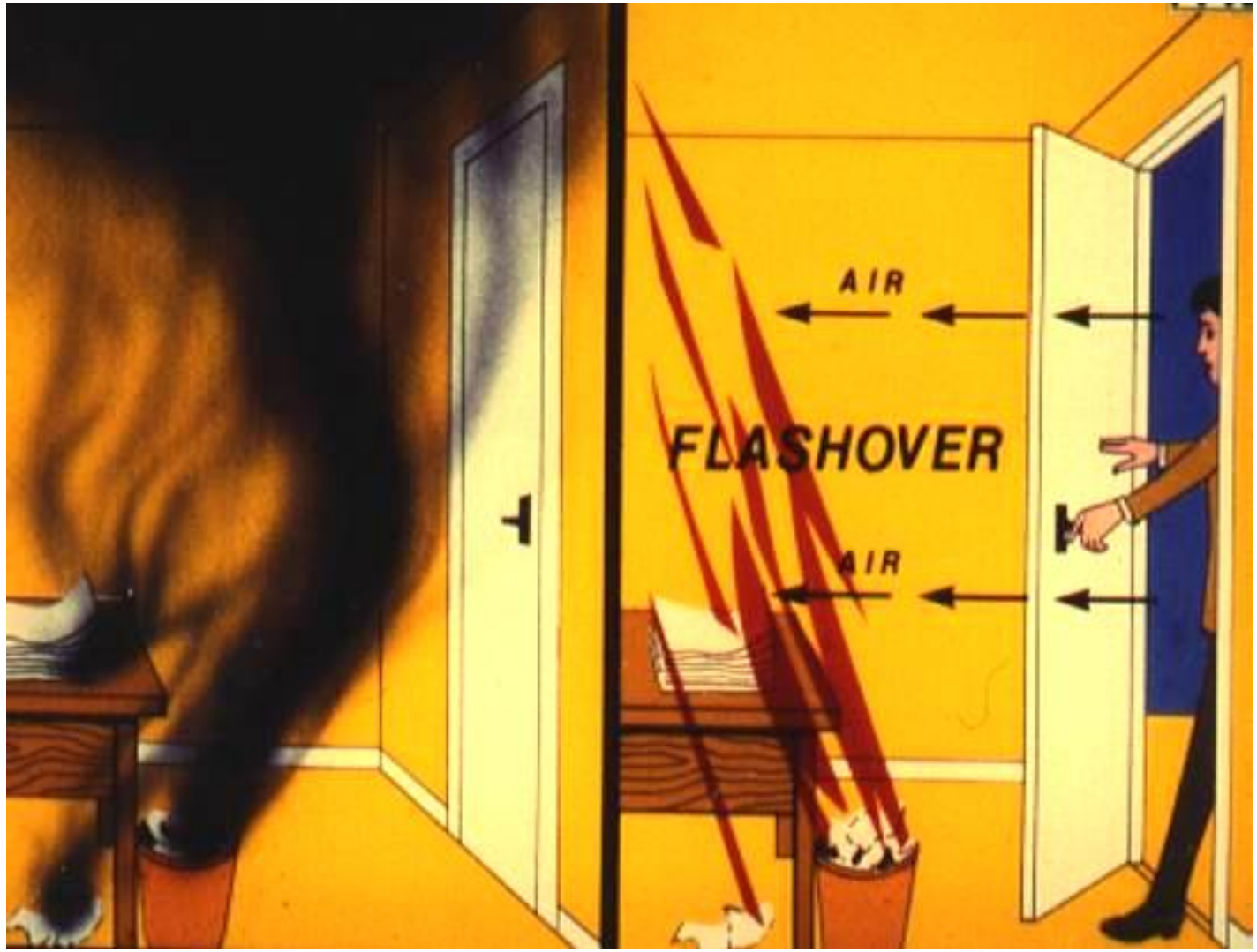
2. At least once every twelve months, maintenance checks shall be made of mechanical parts, the amount and condition of extinguishing agent and expellant, and the condition of the hose, nozzle, and vessel to determine that the fire extinguishers will operate effectively.

What should you do once the fire is extinguished?

1. Check the area closely for any signs of reignition
2. Clean up all unburned and burned materials
3. Report the incident to the proper officials
4. What caused the fire, and determine any safety precautions to prevent future incidents
5. Be careful the fire fighter or persons themselves did not expose to any noxious, toxic or carcinogen products.

Sudden Flash

- Combustible gases formed by the heating of solid materials in an insufficient air environment start the fire by shining with the sudden incoming fresh air.



Sudden Flash

- Combustible gases formed by the heating of solid materials in an insufficient air environment start the fire by shining with the sudden incoming fresh air.

Spread of Fire

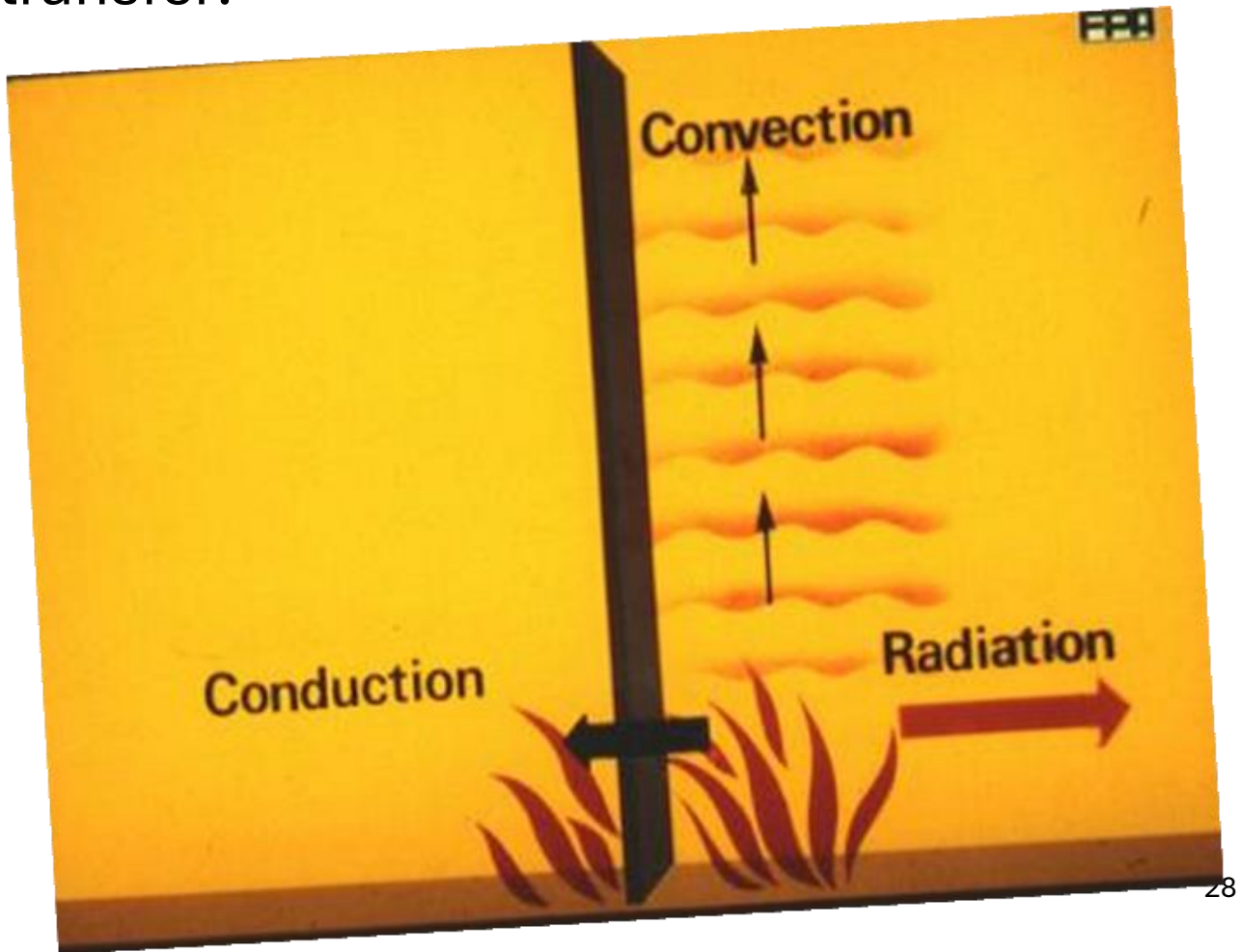
- The spread of fire occurs through heat transfer.

Heat transfer occurs in three ways:

1. **Conduction:** By contact

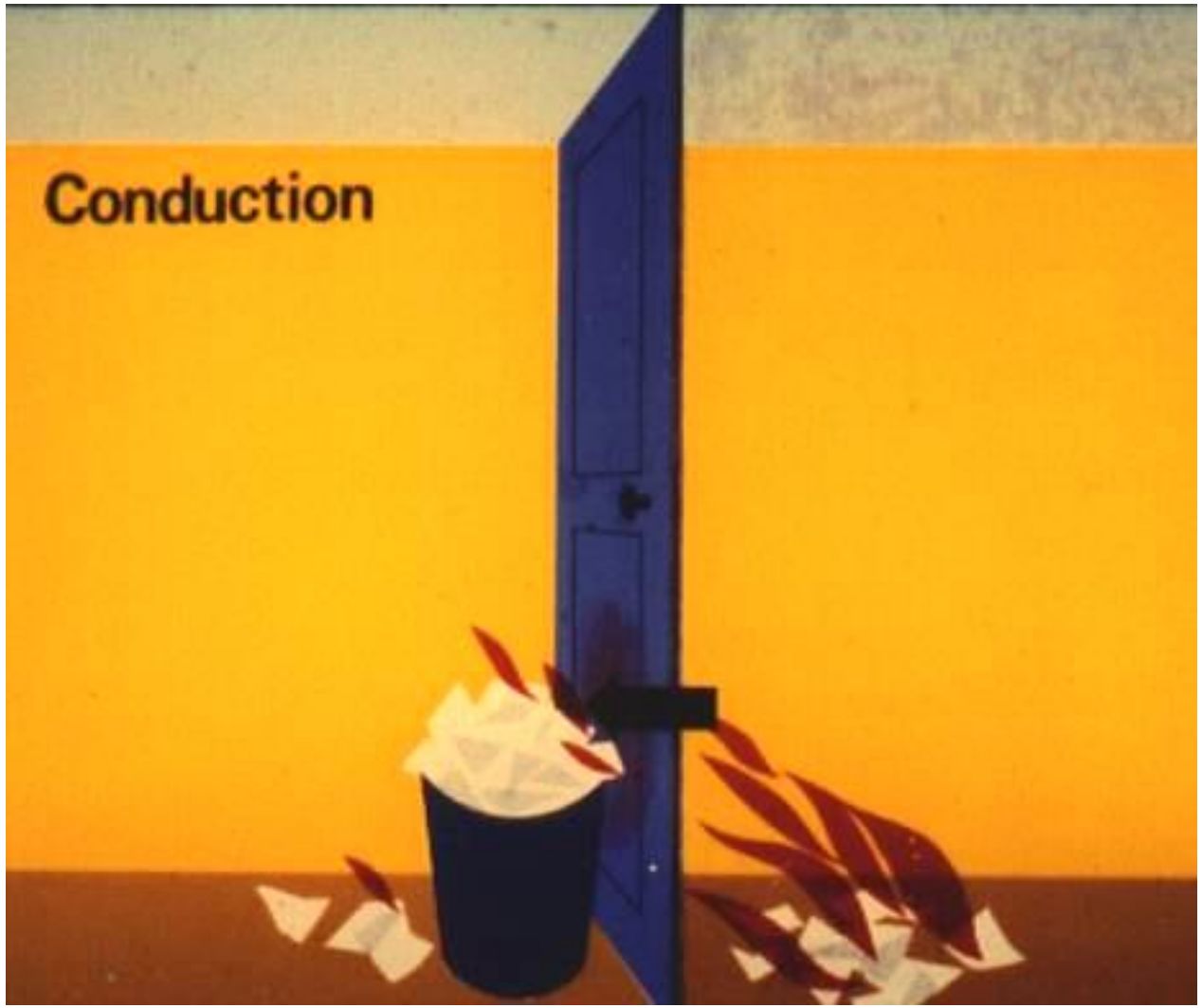
2. **Convection:** By Air

3. **Radiation:** By Radiation



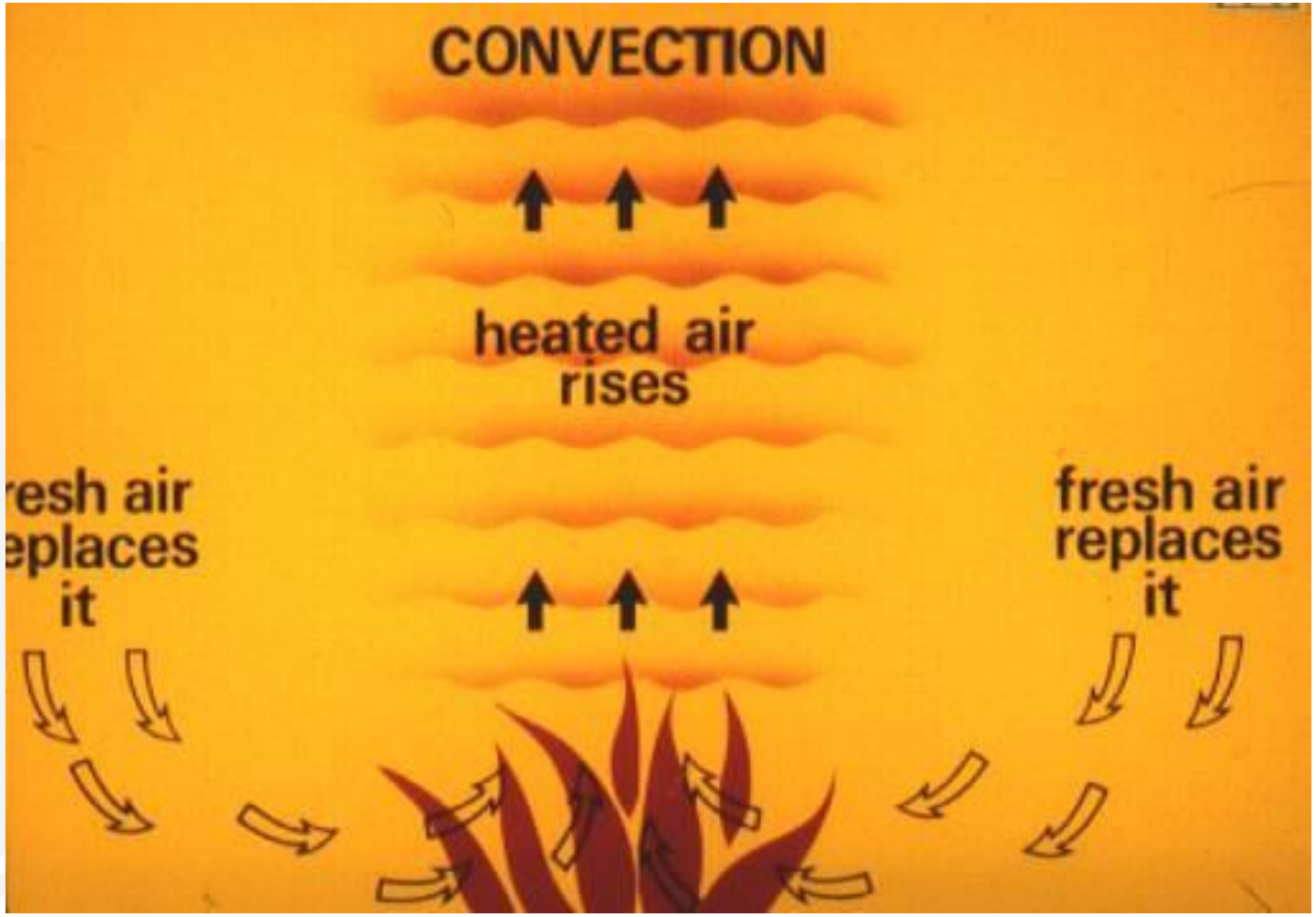
Spread of Fire Conduction

- With the increase in the temperature of surfaces such as walls, ceilings and floors, doors, heat is transmitted to the air and other surfaces in contact.



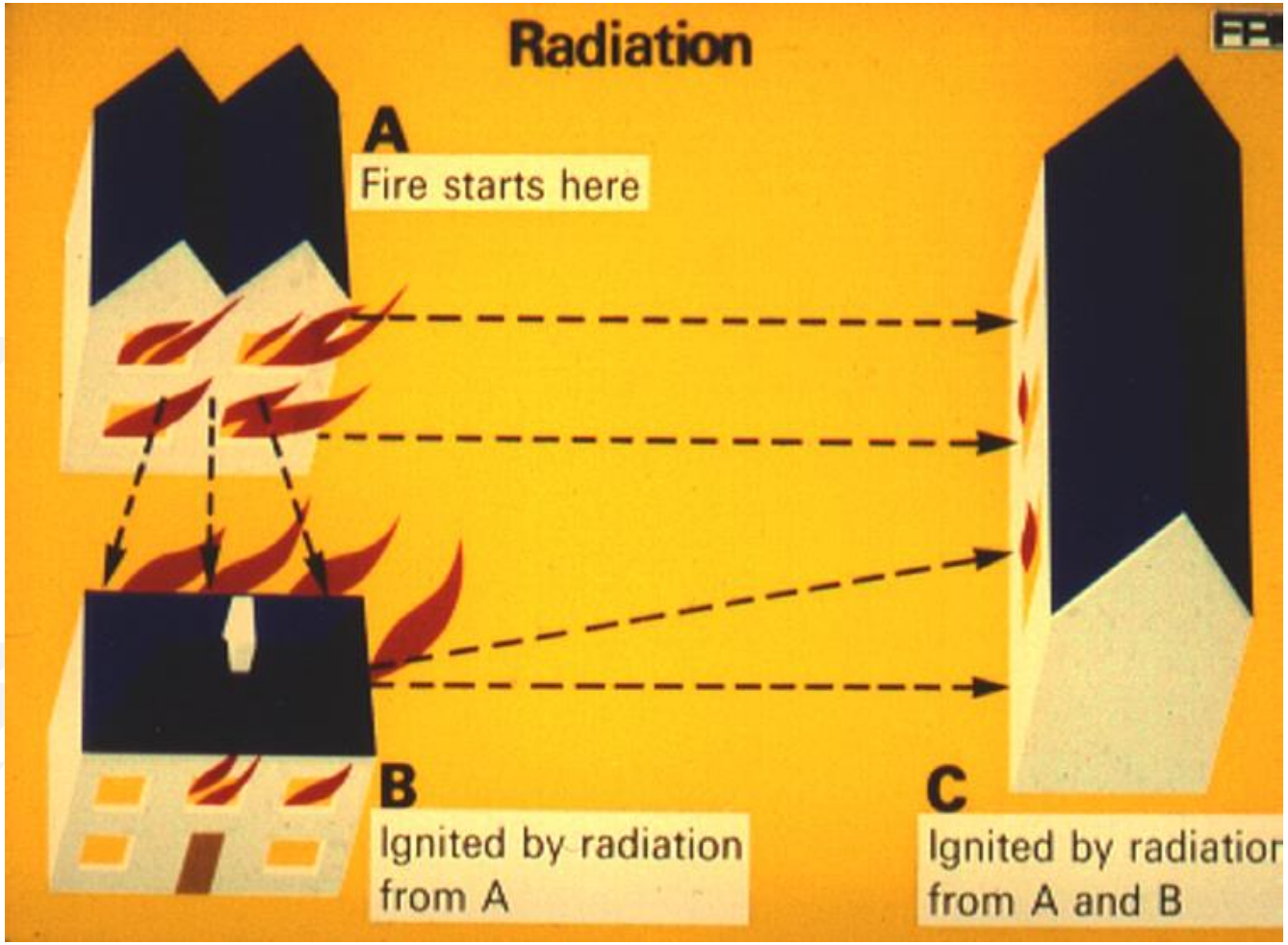
Spread of Fire Convection

- Heated air and hot gases rise, replaced by fresh cold air.
- In this way, the heat spreads rapidly and the fire is fed with the oxygen carried by the air coming into the environment.



Spread of Fire - Radiation

- Heat energy is transmitted as infrared radiation by electromagnetic waves.
- It causes the temperature to increase on the surfaces it hits.

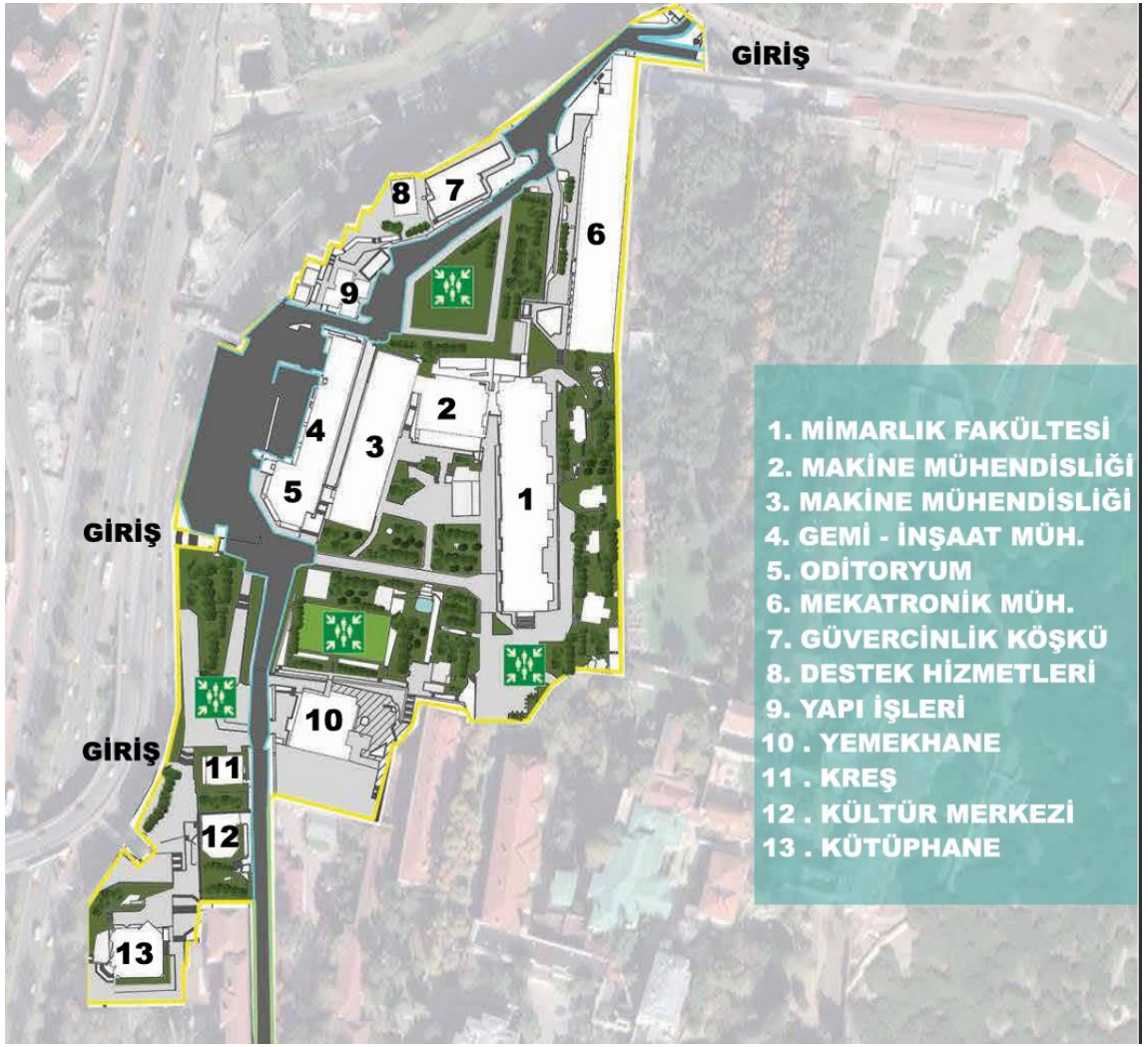


Fire Action Style

1. Alarm Notify
2. First intervention
3. Evacuation
4. Attendance

[YTU Mechanical Faculty emergency action plan]

Yildiz Campus Assembly Points



Fire extinguishing equipment

Fire Class	Combustible Material	Effective Extinguishing Agent
Class (A)	Wood, Coal, Paper, Fabric	Water, Dry Chemical Powder
Class (B)	Fuel, Solvent, Alcohol, Paint	Dry Chemical Powder, Foam, CO_2
Class (C)	Natural Gas, LPG, Acetylene, Oxygen	Dry Chemical Powder, CO_2
Class (D)	Aluminum, Magnesium, Sodium	Dry Chemical Powder

Water Extinguisher

Dry Powder Fire Extinguishers

- They contain ammonium phosphate based chemeical powder. The spraying of the powder is provided by the nitrogen gas pressure inside the device.
- It is effective on (A-B-C) class fires.
- Since dry chemical powders are somewhat abrasive, necessary cleaning should be done as soon as possible after the fire is extinguished.

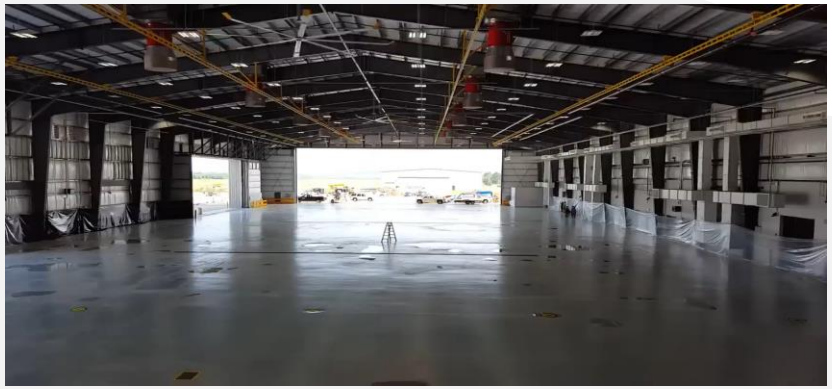
Fire Extinguishers with CO2

- It is filled into steel drawn tubes under high pressure. These devices should always be placed in places away from the sun and heat, and when used indoors, precautions should be taken against oxygen depletion.

- In open field fires and in the wind, CO_2 gas is not very effective by dispersing.

Fire Extinguishers with Halon Gas

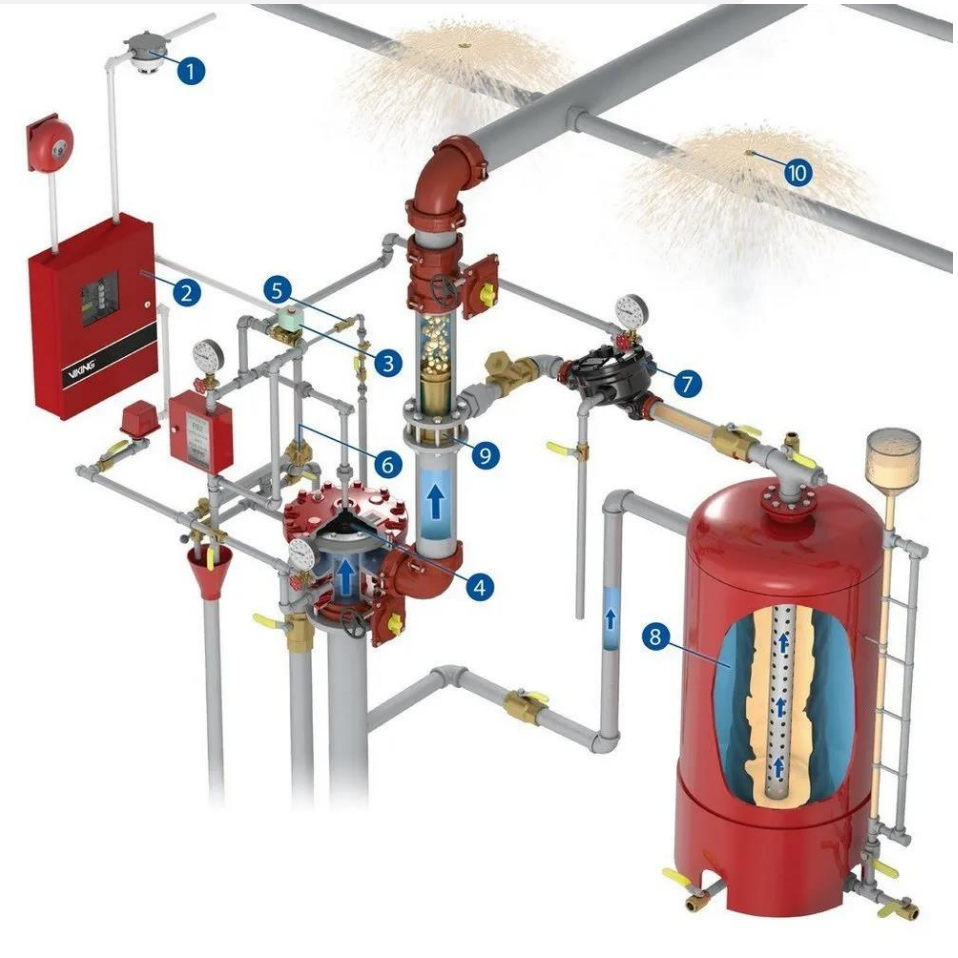
- They are devices that contain halon gas compound. They are gases heavier than air, extinguish the fire by chemical reaction.
- There is no need for any cleaning process after the extinguishing process. It is especially preferred in sensitive machine and electronic device fires.
- It is a colorless and odorless gas, it has toxic properties.
- In the event of a halon gas discharge, that section must be evacuated as quickly as possible.



Aircraft hangar fire alarm test high expansion foam

foaming agent (typically AFFF: Aqueous Film-Forming Foam).

Foam



System Activation: High Expansion Foam Deluge System

- Foam formed by mixing the foaming agent used for fire extinguishing with water under pressure, covers the burning surface and cuts off contact with air.
- It is especially used in fuel tank fires and in large quantities.
- Its use in portable extinguishing devices is limited and not common.

ELIDE FIRE®

AUTOMATIC extinguishing ball

APFE CORP.

DRY POWDER

- A — ✓ USE ON: Wood, Paper and Textiles
- B — ✓ USE ON: Flammable Liquids
- C — ✓ USE ON: Gaseous Fires
- ✓ USE ON: Live Electrical Equipment

FIRE HOSE REEL

- A — ✓ USE ON: Wood, Paper and Textiles
- DO NOT USE ON: Live Electrical Equipment
- B — DO NOT USE ON: Flammable Liquids
- METAL — DO NOT USE ON: Flammable Metal Fires

FOAM SPRAY

- A — ✓ USE ON: Wood, Paper and Textiles
- B — ✓ USE ON: Flammable Liquids
- DO NOT USE ON: Live Electrical Equipment
- METAL — DO NOT USE ON: Flammable Metal Fires

WATER

- A — ✓ USE ON: Wood, Paper and Textiles
- DO NOT USE ON: Live Electrical Equipment
- B — DO NOT USE ON: Flammable Liquids
- METAL — DO NOT USE ON: Flammable Metal Fires

CO_2

- B — ✓ USE ON: Flammable Liquids
- ✓ USE ON: Live Electrical Equipment
- A — DO NOT USE ON: Wood, Paper and Textiles
- DO NOT: Hold horn when operating

Types of fire extinguisher and their uses

WATER

Used on paper, wood, coal, cardboard and other solid fuel fires.

Solid Red

Can be used on:
Class A

FOAM

Used on solid fuel fires as well as flammable liquids.

Cream

Can be used on:
Class A
Class B

POWDER

Used on any kind of fire except for Class F cooking oils.

Blue

Can be used on:
Class A
Class B
Class C
Class D
Electrical

CO2

Used on flammable liquids and electrical fires.

Black

Can be used on:
Class B
Electrical

WET CHEMICAL

Used on cooking oil fires as well as combustible solid materials.

Yellow

Can be used on:
Class A
Class F

Class A

Combustible materials. These include paper, textiles, wood and similar materials.

Class B

Flammable liquids. These include petrol, oil and paint.

Class C

Flammable gases. These include butane and methane.

Class D

Flammable metals. These include potassium and uranium.

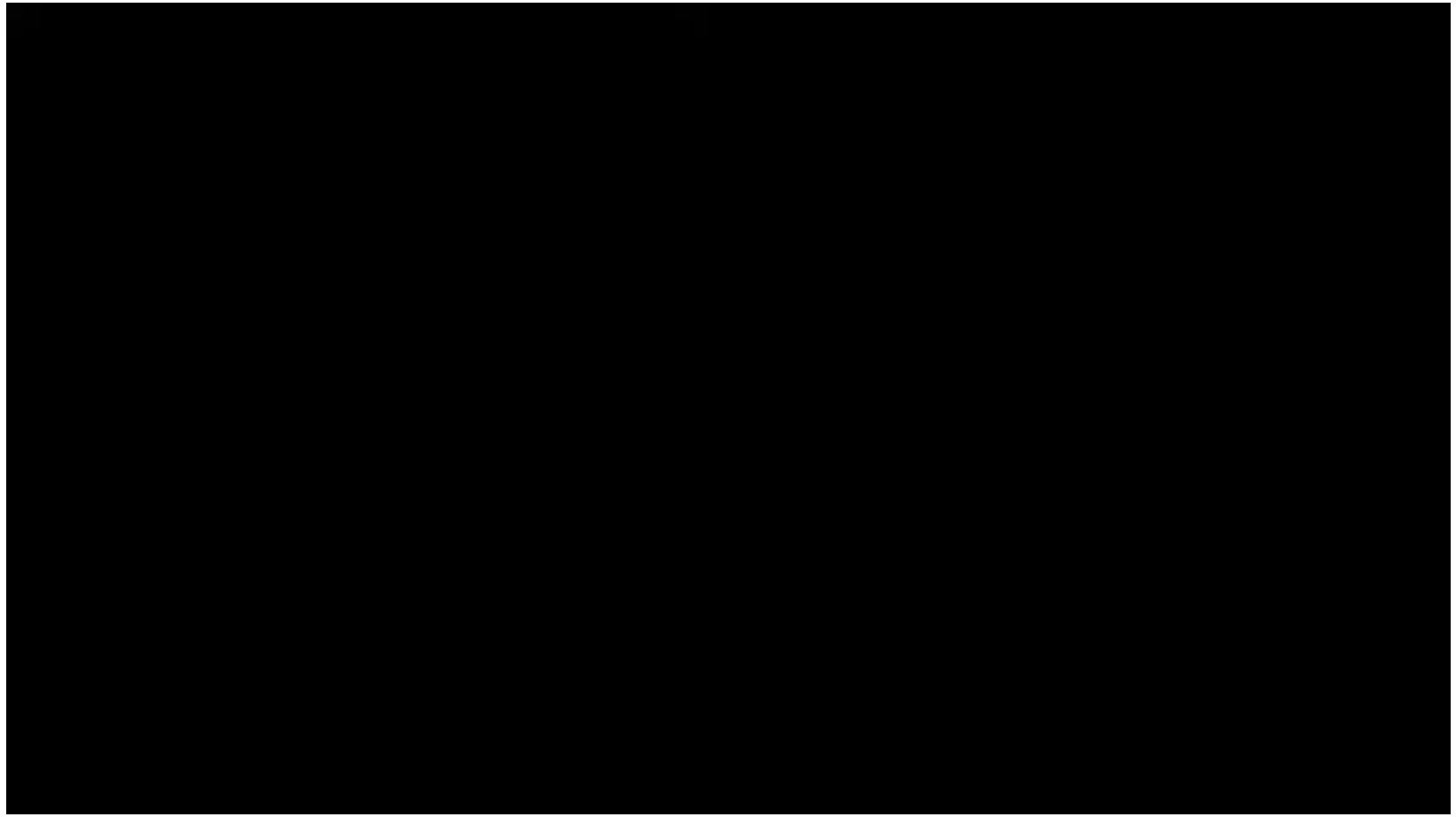
Electrical

Electrical goods. These include appliances in kitchens as well as computers, phones etc.

Class F

Cooking oils. These include chip pan fires and deep fat fryers.

Summary

Fire Extinguisher Training Video - OSHA - Updated for 2023

Fire Safety Training